### Theorem 8.1 (Section Formula - Internal Division)

Let $O$ be the origin. Let $A$ and $B$ be two points. Let $P$ be the point which divides the line segment $AB$ internally in the ratio $m : n$. If $\vec{a}$ and $\vec{b}$ are the position vectors of $A$ and $B$, then the position vector $\overrightarrow{OP}$ of $P$ is given by

$$\overrightarrow{OP} = \frac{n\vec{a} + m\vec{b}}{n+m}.$$

**Proof**

Since $O$ is the origin, $\vec{a}$ and $\vec{b}$ are the position vectors of $A$ and $B$, we have

$$\overrightarrow{OA} = \vec{a} \text{ and } \overrightarrow{OB} = \vec{b}.$$

Let $\overrightarrow{OP} = \vec{r}$.

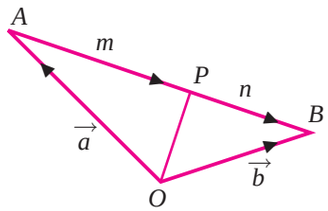


*Fig. 8.26*

Since $P$ divides the line segment $AB$ internally in the ratio $m : n$, we have,

$$\frac{|\overrightarrow{AP}|}{|\overrightarrow{PB}|} = \frac{m}{n}$$

and hence $$n|\overrightarrow{AP}| = m|\overrightarrow{PB}|.$$

But the vectors $\overrightarrow{AP}$ and $\overrightarrow{PB}$ have the same direction. Thus

$$n\overrightarrow{AP} = m\overrightarrow{PB}. \qquad (8.1)$$

But $\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \vec{r} - \vec{a}$ and $\overrightarrow{PB} = \overrightarrow{OB} - \overrightarrow{OP} = \vec{b} - \vec{r}$

Substituting this in (8.1), we get

$$n(\vec{r} - \vec{a}) = m(\vec{b} - \vec{r})$$

and hence

$$(n+m)\vec{r} = n\vec{a} + m\vec{b}.$$

Thus $$\overrightarrow{OP} = \frac{n\vec{a} + m\vec{b}}{n+m}.$$

### Theorem 8.2 Section Formula - External Division (Without proof)

Let $O$ be the origin. Let $A$ and $B$ be two points. Let $P$ be the point which divides the line segment $AB$ externally in the ratio $m : n$. If $\vec{a}$ and $\vec{b}$ are the position vectors of $A$ and $B$, then the position vector $\overrightarrow{OP}$ of $P$ is given by

$$\overrightarrow{OP} = \frac{n\vec{a} - m\vec{b}}{n-m}.$$

**Note 8.1**

By taking $m = n = 1$ in Theorem 8.1, we see that the position vector of the midpoint of the line joining the points $A$ and $B$ is $\frac{\vec{a} + \vec{b}}{2}$, where $\vec{a}$ and $\vec{b}$ are the position vectors of the points $A$ and $B$ respectively.

From the above theorem we can get a condition for three points to be collinear.



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